

COMMAND NEXUS · CIT 140 · WEEK 1

Problem Brief

Meridian Forge Robotics · The Process Below

LEARNER

Craig Temple

ARTIFACT

Problem Brief

STATUS

Working document

SECTION 01

Visible Failure

02:13:07 — MR-17 accepted TEST_K44_START. Source: Incident Log

02:13:09 — MR-17 entered a repeat cycle. Source: Incident Log

02:14:02 — An operator’s stop request returned

The scheduler shows no active test assignment for MR-17.

Test K-44 does not appear in the current test catalog.

Test K-44 was marked retired 186 days before the incident and recorded as removed from the active workflow.

The affected QA cell is currently paused.

Nia recommends preserving evidence before a potential restart of MR-17.

SECTION 02**Verified Facts**

MR-17 accepted TEST_K44_START at 02:13:07, and the incident log records that a repeat cycle began at 02:13:09. Source: Incident Log.

The operator's stop request was accepted by MR-17 but returned ACTIVE_INSTRUCTION_PENDING at 02:14:02, and MR-17 reported that its current test had to finish first. Source: Incident Log / Full Incident Report.

The current scheduler shows no active test assignment for MR-17, and Test K-44 does not appear in the current test catalog. Source: Current Scheduler / Test Catalog.

The maintenance record states that Test K-44 was retired 186 days before the incident and removed from the active workflow. Source: Maintenance Record.

SECTION 03

Assumptions and Open Questions

Assumption: “Repeating” means MR-17 executed Test K-44 more than once or entered a repeat cycle.

Question: What exactly does “repeating” mean in the incident record; multiple completed executions, repeated start attempts, or another behavior?

Question: What potential process, user, service, or parent job generated TEST_K44_START for MR-17 ?

Question: If K-44 was retired and removed from the active workflow, how did TEST_K44_START reach MR-17? Is it possible to query all jobs with Test ID eq 'K-44'?

Question: What does ACTIVE_INSTRUCTION_PENDING specifically mean, and what instruction was pending when the stop request was received?

Question: What does MR-17 mean by “current test must finish first” if the scheduler shows no active assignment? Does that apply to Test K-44 or this previous unknown job?

Question: What is Test k-44?

Question: What is the normal execution time for K-44, and is the two-second interval between TEST_K44_START and the repeat cycle consistent with normal behavior? Has this job ever had issues in the past?

Question: Are there other records showing K-44 assigned to MR-17 or another system around 02:13?

Question: What exactly caused the QA cell to be placed in a “paused” state, and was that pause related to the stop request?

SECTION 04

System Environment Map

SECTION 04A**People and Roles**

Nia: QA supervisor
MR-17: Test robot
Elias: Engineer
Mara: Ops manager

SECTION 04B**Processes and Rules**

The active test scheduler and unknown workflow controls.

SECTION 04C**Information and Data**

Incident Log
Stop command
Current scheduler
Test Catalog
Telemetry
Nia's report
Elia's report
Mara's report

SECTION 04D**Technology and Tools**

MR-17 Robot
legacy_sync.exe
Interface (used for stop command)

SECTION 04E**Environment and Constraints**

QA cell currently being paused affects workflow/production.
No deadline has been given.
No permissions of user have been validated.
No cost documented as still in investigation.

SECTION 05

Investigation Boundary

Inside the Investigation

The team will now investigate preserving evidence for exploring restart.

Outside for Now

Restart of MR-17
Scheduler changes of jobs
User permissions
Production changes for workflow

SECTION 06

Organizational Impact

Current workflow paused.
User resources for investigation (Nia/Elias/Mara)

SECTION 07

Problem Statement

Gap: MR-17 running a job that was supposed to be removed.
Impact: QA cell is currently paused due to error running job.

SECTION 08

Next Investigation Step

Trace origin of Test case K-44 command. Source could be in the legacy_sync.exe logs. This would answer what user or process kicked off this retired job.

SHARED ARTIFACT

Working Document Link

[Not entered](#)

WEEK 1 STOPPING RULE**Define before designing.**

This Problem Brief records the problem that requires investigation. It does not recommend a solution.